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Elektornika Analog Latihan Soal BAB 9

1. Calculate gm0 for a JFET having device parameters IDSS = 15 mA and VP = -5 V

   $gm0 = \frac{2IDSS}{|Vp|} = \frac{2(15)}{|-5|} =$ **6 mS**

2. Determine the pinch-off voltage of a JFET with gm0 = 10 mS and IDSS = 12 mA.

   $gm0 = \frac{2IDSS}{|Vp|}$ $|VP| = \frac{2IDSS}{gm0} = \frac{2(12)}{10} =$ **VP = 2.4 V**

3. For a JFET having device parameters gm0 = 5 mS and VP = -3.5 V, what is the device current at VGS = 0 V?

   $gm0 = \frac{2IDSS}{|Vp|}$ $IDSS = \frac{(gm0)(|Vp|)}{2} = \frac{(5)(3.5)}{2} =$ **8.75 mA**

4. Calculate the value of gm for a JFET (IDSS = 12 mA, VP = -3 V) at a bias point of VGS = -1 V.

   $gm = gm0\ 1 - \frac{VGSQ}{Vp} = \frac{2(12)}{-3}\ 1 - \frac{-1}{-3} =$ **5.33 mS**

5. For a JFET having gm = 6 mS at VGSQ = -1 V, what is the value of IDSS if VP = -2.5 V?

   $gm = \frac{2IDSS}{|Vp|}\ 1 - \frac{VGSQ}{Vp}$

   $mS = \frac{2IDSS}{25}\ 1 - \frac{-1}{-25} =$ **IDSS = 2 Ms**

6. A JFET (IDSS = 10 mA, VP = -5 V) is biased at ID = IDSS /4. What is the value of gm at that bias point?

   $gm = gm0\sqrt{\frac{ID}{IDSS}} = \frac{2IDSS}{|Vp|}\sqrt{\frac{IDSS/4}{IDSS}}\sqrt{1 - \frac{2(10)}{5V}} = \frac{20\ mA}{5V}\ \frac{1}{2} =$ **2 mS**

7. Determine the value of gm for a JFET (IDSS = 8 mA, VP = -5 V) when biased at VGSQ = VP /4.

   $gm0 = \frac{2IDSS}{|Vp|} = \frac{2(8)}{5V} =$ 3.2 mS

   $gm = gm0\ 1 - \frac{VGSQ}{Vp} =$ 3.2 mS $1 - \frac{Vp/4}{Vp} =$ 3.2 mS $1 - \frac{1}{4} =$ 3.2 mS $\frac{3}{4}$ **2.4 mS**

8. A specification shect provides the following data (at a listed drain-source current) y_{fs} = 4.5 mS, y_{os} = 25 µS. At the listed drain-source current, determine : a. g_m = g_{fs} = **4.5 mS**

   b. $r_d = \frac{1}{g_{os}} = \frac{1}{25\mu S} =$ **40$k\Omega$**

9. For a JFET having specified values of y*fs*=4.5 mS and y*os*=25 µS, determine the device output impedance, Z*o* (FET), and device ideal voltage gain, A*v* (FET).

   $g_m = y_{fs} = 4.5\ mS$

   $r_d = \frac{1}{y_{os}} = \frac{1}{25\mu S} = 40k\Omega$

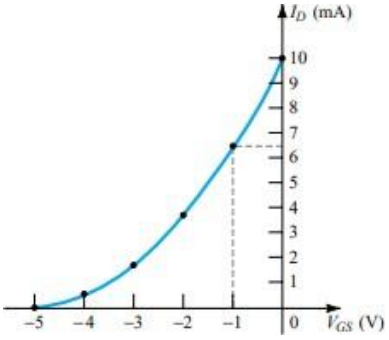
   $Z_O = r_d =$ **40$k\Omega$**

$A_v(FET) = -g_m r_d = -(4.5\ mS)(40k\Omega) = \mathbf{-180}$

10. If a JFET having a specified value of $r_d = 100k\Omega$ has an ideal voltage gain of $A_v(FET) = -200$, *what is the value of* g_m.

$$A_v = -g_m r_d \Rightarrow g_m = \frac{-A_v}{r_d} = -\frac{(-200)}{(100\ \text{k}\Omega)} = \mathbf{2\ mS}$$

11. Using the transfer characteristic of Fig. 9.52 :



a. What is the value of g_{mo}?

$$g_{mo} = \frac{2I_{DSS}}{|V_p|} = \frac{2(10\ mA)}{5V} = \mathbf{4\ mS}$$

b. Determine g_m at $V_{GS} = -1.5\ V$ graphically.

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{9\ mA - 7\ mA}{0.85\ V - 0.3\ V} = \mathbf{3.64\ mS}$$

c. What is the value of g_m at $V_{GSQ} = -1.5\ V$ using Eq. (9.6)? Compare with the solusion to part (b).

$$g_m = g_{mo}\left(1 - \frac{V_{GSQ}}{V_P}\right) = 4\,mS\left(1 - \frac{-1.5\ V}{-5\ V}\right) = \mathbf{2.8\ mS}$$

d. Graphically determine g_m at $V_{GS} = -2.5\ V$

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{8\ mA - 5\ mA}{2.5\ V - 1.5\ V} = \mathbf{3\ mS}$$

e. What is the value of g_m at $V_{GSQ} = -2.5\ V$ using Eq. (9.6)? Compare with the solusion to part (d)

$$g_m = g_{mo}\left(1 - \frac{V_{GSQ}}{V_P}\right) = 4\ mS\left(1 - \frac{-2.5\ V}{-5\ V}\right) = \underline{\mathbf{2\ mS}}$$

12. Using the drain characteristic of Fig. 9.53 :

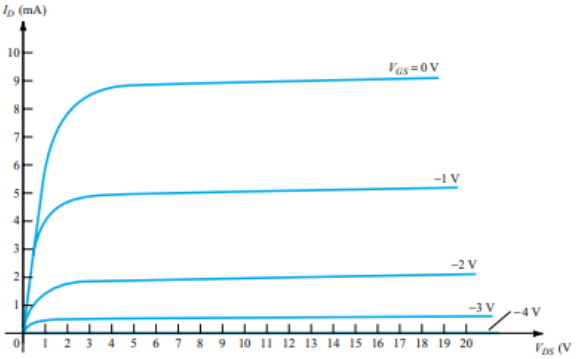


Figure 9.53 JFET drain characteristic for Problem 12

a. What is the value of r*d* for VGS = 0 V

$$r_d = \left.\frac{\Delta V_{DS}}{\Delta I_D}\right|_{V_{GS}\ \text{constant}} = \frac{(15\ \text{V} - 5\ \text{V})}{(9.1\ \text{mA} - 8.8\ \text{mA})} = \frac{10\ \text{V}}{0.3\ \text{mA}} = \mathbf{33.33\ k\Omega}$$

b. What is the value of g_{m0} *at* $V_{DS} = 10\ V$?

At $V_{DS} = 10$ V, $I_D = 9$ mA on $V_{GS} = 0$ V curve

$$\therefore g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(9\ \text{mA})}{4\ \text{V}} = \mathbf{4.5\ mS}$$

13. For a 2N4220 n-channel JFET (y_{fs}(minimum) = 750 µS, y_{Os}(maximum) = 10 µS) :

a. What is the value of g_m?

$g_m = g_{fs} = 750\ \mu S = \mathbf{0.75\ mS}$

b. What is the value of r_d? $r_d = \frac{1}{g_{os}} = \frac{1}{10\ \mu S} = \mathbf{100k\Omega}$

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4
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a. Plot g_m *vs*, V_{GS} for an n channel JFET with $I_{DSS} = 8\ mA$ and $V_p = -6\ V$

$g_m(V_{GS} = -6\ V) = 0$, $g_{m}(V_{GS} = 0\ V) = g_{mo} = \frac{2I_{DSS}}{|V_p|} = \frac{2(8\ mA)}{6\ V} = \mathbf{2.7\ mS}$

b. Plot g_m *vs*, I_D for the same n-channel JFET as part (a).

$g_m(I_D = 0\ mA) = 0$, $g_m(I_D = I_{DSS} = 8\ mA) = g_{mO} = \frac{2I_{DSS}}{|V_p|} = \frac{2(8\ mA)}{6\ V} = \mathbf{2.7\ mS}$

15. Sketch the ac equivalent model for a JFET if $y_{fs} = 5.6\ mS$ and $y_{Os} = 15\mu S$.

$g_m = g_{fs} = 5.6$ mS, rd = $\frac{1}{g_{Os}} = \frac{1}{15\ \mu S} = \mathbf{66.67\ k\Omega}$

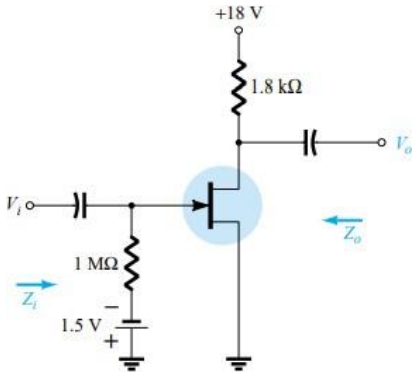
16. Sketch the ac equivalent model for a JFET if $I_{DSS} = 10\ mA$ and $V_p = -4\ V$, $V_{GSQ} = -2\ V$, and $y_{Os} = 25\mu S$.

$$g_m = \frac{2I_{DSS}}{|V_p|}\left(1 - \frac{V_{GSQ}}{V_P}\right) = \frac{2(10\ mA)}{4\ V}\left(1 - \frac{-2\ V}{-4\ V}\right) = \mathbf{2.5\ mS}$$

$$r_d = \frac{1}{g_{os}} = \frac{1}{25\ \mu S} = \mathbf{40k\Omega}$$

17. Determine $Z_i, Z_0 and A_v$ for the network of Fig 9.54 if $I_{DSS} = 10\ mA$, $V_p = -4\ V$, and $r_d = 40k\Omega$

18. Determine $Z_i, Z_0 and A_v$ for the network of Fig 9.54 if $I_{DSS} = 10\ mA$, $V_p = -4\ V$, and $r_d = 40k\Omega$



Graphically, $V_{GSQ} = -1.5\ V$

$$g_m = \frac{2I_{DSS}}{|V_p|}\left(1 - \frac{V_{GSQ}}{V_P}\right) = \frac{2(10\ mA)}{4\ V}\left(1 - \frac{-1.5\ V}{-4\ V}\right) = 3.12\ mS$$

$Z_i = R_G = \mathbf{1\ M\Omega}$

$Z_O = R_D \| r_d = 1.8\ \text{k}\Omega \| 40\ \text{k}\Omega = \mathbf{1.72\ k\Omega}$

$A_v = -g_m(R_D \| r_d) = -(3.12\ mS)(1.72\ k\Omega) = \mathbf{-5.36}$

19. Determine $Z_i, Z_O and A_v$ for the network of Fig 9.54 if $I_{DSS} = 12\ mA, V_p = -6\ V,$ and $r_d = 40\mu S.$

20. Determine $Z_i, Z_O and A_v$ for the network of Fig 9.55 if $y_{fs} = 3000\mu S$ and $y_{os} = 50\mu s.$

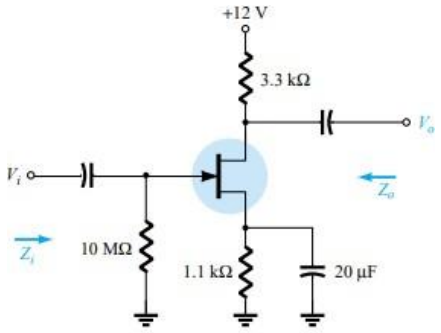


Figure 9.55 Problems 19, 21, and 46

$g_m = y_{fs} = 3000\ \mu S = 3\ \text{mS}$

$$r_d = \frac{1}{y_{os}} = \frac{1}{50\ \mu S} = 20\ \text{k}\Omega$$

$Z_i = R_G = \mathbf{10\ M\Omega}$

$Z_o = r_d \| R_D = 20\ \text{k}\Omega \| 3.3\ \text{k}\Omega = \mathbf{2.83\ k\Omega}$

$A_v = -g_m(r_d \| R_D)$

$= -(3\ \text{mS})(2.83\ \text{k}\Omega)$

$= \mathbf{-8.49}$

21. Determine $Z_i, Z_O and A_v$ for the network of Fig 9.56 if $I_{DSS} = 6\ mA, V_p = -6\ V,$ and $r_d = 40\mu S.$

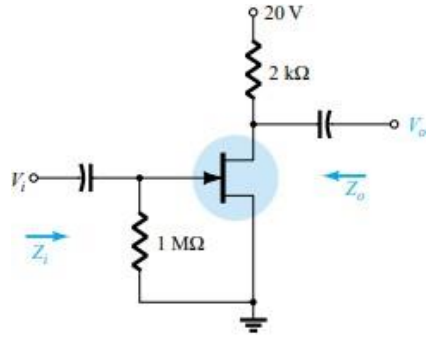


Figure 9.56 Self-bias configuration for Problems 20 and 47

$$V_{GS_Q} = 0\ \text{V}, g_m = g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(6\ \text{mA})}{6\ \text{V}} = 2\ \text{mS}, r_d = \frac{1}{y_{os}} = \frac{1}{40\ \mu S} = 25\ \text{k}\Omega$$

$Z_i = \mathbf{1\ M\Omega}$

$Z_o = r_d \| R_D = 25\ \text{k}\Omega \| 2\ \text{k}\Omega = \mathbf{1.852\ k\Omega}$

$A_v = -g_m(r_d \| R_D) = -(2\ \text{mS})(1.852\ \text{k}\Omega) \cong \mathbf{-3.7}$

22. Determine $Z_i, Z_O and A_v$ for the network of Fig 9.55 if the $20 - \mu F$ capasitor is removed and the parameters of the network are the same as in Problem 19. Compare result with those of Problem 19.

$g_m = 3\ \text{mS}, r_d = 20\ \text{k}\Omega$

$Z_i = \mathbf{10\ M\Omega}$

$$Z_o = \frac{R_D}{1 + g_m R_S + \dfrac{R_D + R_S}{r_d}} = \frac{3.3\ \text{k}\Omega}{1 + (3\ \text{mS})(1.1\ \text{k}\Omega) + \dfrac{3.3\ \text{k}\Omega + 1.1\ \text{k}\Omega}{20\ \text{k}\Omega}}$$

$$= \frac{3.3\ \text{k}\Omega}{1 + 3.3 + 0.22} = \frac{3.3\ \text{k}\Omega}{4.52} = \mathbf{730\ \Omega}$$

$$A_v = \frac{-g_m R_D}{1 + g_m R_S + \dfrac{R_D + R_S}{r_d}} = \frac{-(3\ \text{mS})(3.3\ \text{k}\Omega)}{1 + (3\ \text{mS})(1.1\ \text{k}\Omega) + \dfrac{3.3\ \text{k}\Omega + 1.1\ \text{k}\Omega}{20\ \text{k}\Omega}}$$

$$= \frac{-9.9}{1 + 3.3 + 0.22} = -\frac{9.9}{4.52} = \mathbf{-2.19}$$

23. Repeat Problem 19 if $y_{os} = 10\mu S$. Compare the result to those of Problem 19.

$g_m = y_{fs} = 3000\ \mu\text{S} = 3\ \text{mS}$

$$r_d = \frac{1}{y_{os}} = \frac{1}{10\ \mu\text{S}} = 100\ \text{k}\Omega$$

$Z_i = R_G = \mathbf{10\ M\Omega}$ (the same)

$Z_o = r_d \parallel R_D = 100\ \text{k}\Omega \parallel 3.3\ \text{k}\Omega = \mathbf{3.195\ k\Omega}$ (higher)

$A_v = -g_m(r_d \parallel R_D)$

$= -(3\ \text{mS})(3.195\ \text{k}\Omega)$

$= \mathbf{-9.59}$ (higher)

24. Determine Z_i, Z_o *and* V_o for the network of Fig 9.57 if $V_i = 20\ mV$.

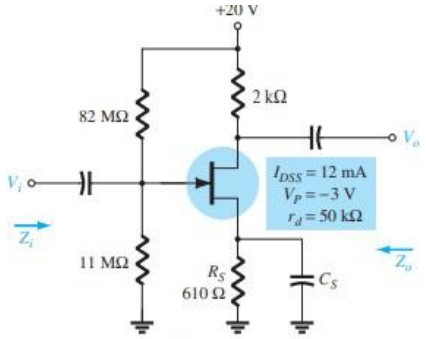


FIG. 8.76

$-0.95\ V$

$V_{GSQ} = -0.95\ V$

$$g_m = \frac{2I_{DSS}}{|V_p|}\left(1 - \frac{V_{GSQ}}{V_P}\right) = \frac{2(12\ mA)}{3\ V}\left(1 - \frac{}{-3\ V}\right) = 5.47\ mS$$

$Z_i = 82\ M\Omega \| 11\ M\Omega = \mathbf{9.7\ M\Omega}$

$Z_O = r_d \| R_D = 50\ \text{k}\Omega\ \| 2\ \text{k}\Omega = \mathbf{1.92\ k\Omega}$

$A_v = -g_m(r_d \| R_D) = -(5.47\ mS)(1.92\ K\Omega) = \mathbf{-10.5}$

$V_o = A_v V_i = (-10.5)(20\ mV) = \mathbf{-210\ mV}$

25. Determine Z_i, Z_o *and* V_o for the network of Fig 9.57 if $V_i = 20\ mV$ and the capasitor C_s is removed.

V_{GS_Q} = −0.95 V (as before), g_m = 5.47 mS (as before)

Z_i = **9.7 MΩ** as before

$$Z_o = \frac{R_D}{1 + g_m R_S + \dfrac{R_D + R_S}{r_d}}$$

but $r_d \geq 10(R_D + R_S)$

$$\therefore Z_o = \frac{R_D}{1 + g_m R_S} = \frac{2\ \text{k}\Omega}{1 + (5.47\ \text{mS})(0.61\ \text{k}\Omega)} = \frac{2\ \text{k}\Omega}{1 + 3.337} = \frac{2\ \text{k}\Omega}{4.337}$$

= **461.1 Ω**

$$A_v = \frac{-g_m R_D}{1 + g_m R_S} \quad \text{since } r_d \geq 10(R_D + R_S)$$

$$= \frac{-(5.47\ \text{mS})(2\ \text{k}\Omega)}{4.337\ \text{(from above)}} = -\frac{10.94}{4.337} = \textbf{−2.52}\ \text{(a big reduction)}$$

$V_o = A_v V_i = (-2.52)(20\ \text{mV}) =$ **−50.40 mV** (compared to −214.4 mV earlier)

26. Repeat Problem 23 if $r_d = 20k\Omega$ and compare results.

V_{GSQ} = −0.95 V, g_m = 5.47 mS

Z_i (the same) = **9.7 MΩ**

Z_o (reduced) = $r_d \| R_D$ = 20 kΩ || 2 kΩ = **1.82 kΩ**

A_v (reduced) = $-g_m(r_d \| R_D)$ = −(5.47 mS)(1.82 kΩ) = **−9.94**

V_o (reduced) = $A_v V_i$ = (−9.94)(20 mV) = **−198.8 mV**

27. Repeat Problem 24 if rd = 20 kΩ and compare results.

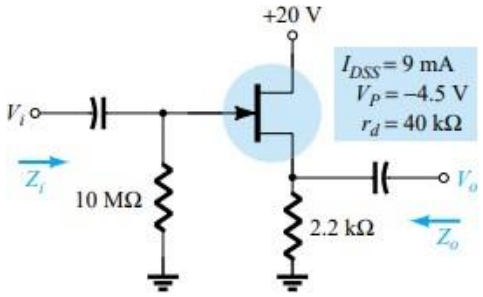
➔ VGSQ = -0.95 V. gm = 5.47 mS

Zi = **9.7 MΩ**

Zo = $\frac{2\ k\Omega}{1+333+013} = \frac{2\ k\Omega}{446}$ = **448.4 Ω**

Av = $\frac{-1094}{1+333+013} = \frac{-1094}{446}$ = **-2.45**

28. Determine Zi, Zo, and Av for the network of Fig. 9.58.



➔ VGSQ = -2.85 V. gm = $\frac{2IDSS}{Vp}\left(1-\frac{VGSQ}{Vp}\right) = \frac{2\,(9\ mA)}{45}\left(1-\frac{-285\ V}{45\ V}\right) = 1.47$ mS

Zi = RG = **10 MΩ**

Zo = rd || Rs || 1/gm = 40 kΩ || 2.2 kΩ || 1/1.47 ms (680.27 Ω) = **512.9 Ω**

Av = $\frac{gm\,(rd\|Rs)}{1+gm\,(rd\|Rs)} = \frac{3065}{1+3065}$ = **0.754**

29. Repeat Problem 27 if rd = 20 kΩ.

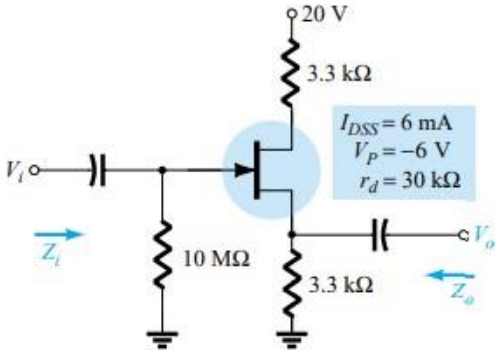
➔ VGSQ = -2.85 V. gm = 1.47 mS

Zi = **10 MΩ**

Zo = rd || Rs || 1/gm = 20 kΩ || 2.2 kΩ || 680.27 Ω = **506.4 Ω**

Av = $\frac{gm\,(rd\|Rs)}{1+gm\,(rd\|Rs)} = \frac{2914}{1+2914}$ = **0.754**

30. Determine Zi, Zo, and Av for the network of Fig. 9.59.



➔ VGSQ = -2.85 V. gm = $\frac{2IDSS}{Vp}\left(1-\frac{VGSQ}{Vp}\right) = \frac{2(6\,mA)}{-6\,V}\left(1-\frac{-38\,V}{6\,V}\right) = \frac{0.733}{-6\,V}$ mS

Zi = RG = **10 MΩ.**

rd' = rd + RD = 30 kΩ + 3.3 kΩ = 33.3 kΩ

gm' = $\frac{gm\,rd}{rd+RD} = \frac{(0733\,mS)(30\,k\Omega)}{30\,k\Omega+33\,k\Omega} = \frac{2199}{333\,k\Omega}$ = 0.66 mS

Zo = rd' || Rs || 1/gm' = 33.3 kΩ || 3.3 kΩ || 1/0.66 mS = **1 kΩ**

Av = $\frac{gm'(rd'\|Rs)}{1+gm'(rd'\|Rs)} = \frac{198}{1+198}$ = **0.66**

31. Determine Zi, Zo, and Vo for the network of Fig. 9.60 if Vi = 0.1 mV.

VGSQ = -1.75 V. gm = 2.14 mS

rd ≥ 10RD. #Zi ≈ Rs || 1/gm = 1.5 kΩ || ½.14 mS = **356.3 Ω**

rd ≥ 10RD. #Zo ≈ RD = **3.3 kΩ**

rd ≥ 10RD. #Av ≈ gmRD = 92.14 mS)(3.3 kΩ) = **7.06**

Vo = AvVi = (7.06)(4 mV) = **28.24 mV**

32. Repeat Problem 30 if rd = 25 kΩ

VGSQ = -1.75 V. gm = $\frac{2IDSS}{Vp}\left(1-\frac{VGSQ}{Vp}\right) = \frac{2(8\,mA)}{-28\,V}\left(1-\frac{-175\,V}{28}\right) = 2.14$ mS

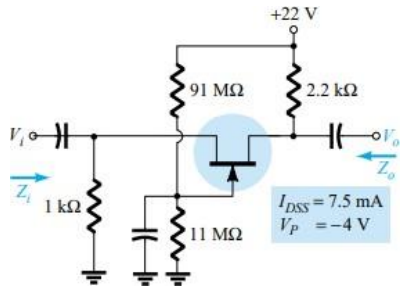
Zi = Rs || $\frac{rd+RD}{1+gm\,rd}$ = 1.5 kΩ || $\frac{283\,k\Omega}{545}$ = **386.1 Ω**

Zo = RD || rd = 3.3 kΩ || 25 kΩ = **2.92 kΩ**

Av = $\frac{gm\,RD+RD/rd}{1+RD/rd} = \frac{7194}{7132}$ = 6.36

Vo = AvVi = (6.36)(4 mV) = **25.44 Mv**

33. Determine Zi, Zo, and Av for the network of Fig. 9.61 if rd = 33 kΩ.



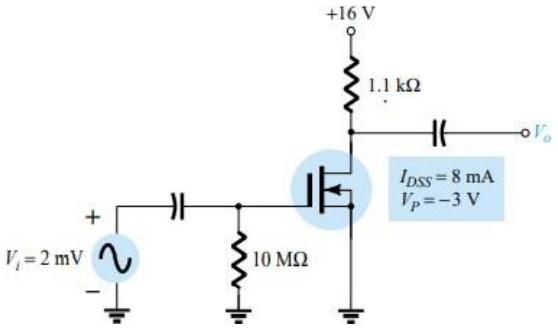
VGSQ ≈ -1.2 V, gm = 2.63 Ms

rd ≥ 10RD #Zi ≈ Rs || 1/gm = 1 kΩ || 1/2.63 mS = **275.5 Ω**

Zi ≈ RD = **2.2 kΩ**

Av ≈ gmRD = (2.63 mS)(2.2 kΩ) = **5.79**

34. Determine Vo for the network of Fig. 9.62 if y*os* = 20 µS.



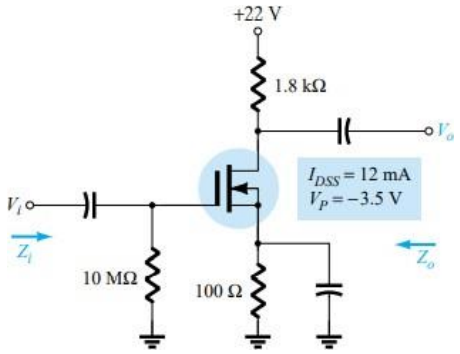
rd = $\frac{1}{yos} = \frac{1}{20\,\mu S}$ = 50 kΩ, VGSQ = 0V

gm = gm0 = $\frac{2IDSS}{Vp} = \frac{2(8)}{3}$ = 5.33 mS

Av = -gmRD = -(5.33 mS)(1.1 kΩ) = -5.863

Vo = AvVi = (-5.863)(2 mV) = **11.73 mV**

35. Determine Zi, Zo, and Av for the network of Fig. 9.63 if rd = 60 kΩ.



VGSQ ≈ -0.75 V, gm = 2.4 Ms

Zi = **10 MΩ**

rd ≥ 10RD #Zo ≈ RD = **1.8 kΩ**

rd ≥ 10RD # Av ≈ -gmRD = -(5.4 mS)(1.8 kΩ) = **-9.72**

36. Repeat Problem 34 if rd = 25 kΩ

Zi = **10 MΩ**

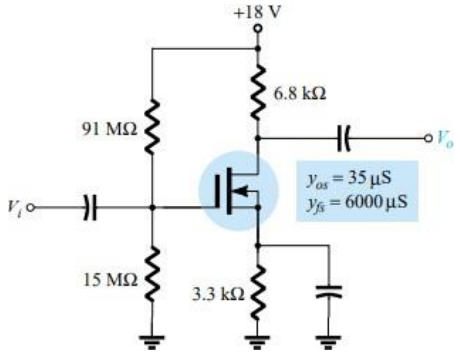
Zo = rd || RD = 25 kΩ || 1.8 kΩ = **1.68 kΩ**

Av = -gm(rd||RD)

gm = $\frac{2IDSS}{Vp}\left(1 - \frac{VGSQ}{Vp}\right) = \frac{2(12\,mA)}{35}\left(1 - \frac{-0.75\,V}{-35\,V}\right)$ = 5.4 mS

Av = -(5.4 mS)(1.68 kΩ) = **-9.07**

37. Determine Vo for the network of Fig. 9.64 if V*i* = 4 mV.



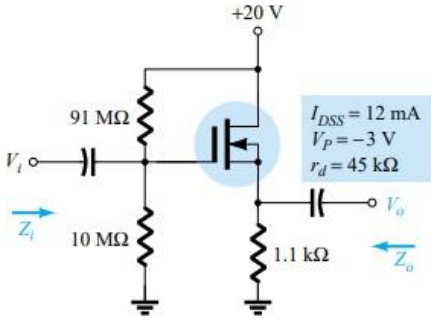
gm = yfs = 6000 µS = 6 mS

rd = $\frac{1}{yos} = \frac{1}{35\,\mu S}$ = 28.57 kΩ

rd ≥ 10RD #Av = -gm(rd||RD)= 1(6 mS)(28.57 kΩ||6.8 kΩ) = **-32.94**

Vo = AvVi = (-32.94)(4 mV) = **-131.76 Mv**

38. Determine Zi, Zo, and Av for the network of Fig. 9.65.



Zi = 10 MΩ || 91 MΩ = **9 MΩ**

gm = $\frac{2IDSS}{Vp}\left(1 - \frac{VGSQ}{Vp}\right) = \frac{2(12\ mA)}{3\ V}\left(1 - \frac{-145\ V}{-3\ V}\right)$ = -4.13 mS

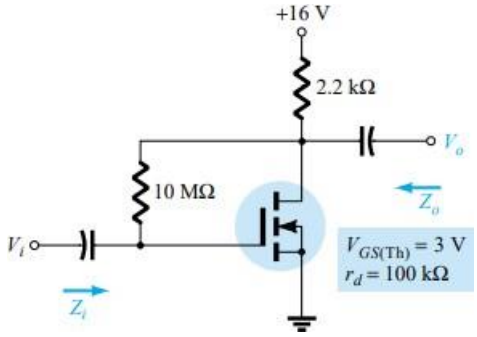
Zo = rd || Rs || 1/gm = 45 kΩ || 1.1 kΩ || ¼.13 mS = **197.6 Ω**

Av = $\frac{gm\,(rd\|Rs)}{1+gm\,(rd\|Rs)} = \frac{4436}{1+4436}$ = **0.816**

39. Determine gm for a MOSFET if VGS(Th) = 3 V and it is biased at VGSQ = 8 V. Assume k 0.3 x 10^{-3}

gm = 2k (VGSQ - VGS(Th)) = 2(0.3 x 10^{-3}) (8V – 3V) = **3 mS**

40. Determine Zi, Zo, and Av for the amplifier of Fig. 9.66 if k = 0.3 x 10^{-3}



VGSQ = 6.7 V

gm = 2k(VGSQ – VT) = 2(0.3 x 10^{-3})(6,7 V – 3 V) = 2.22 mS

Zi = $\frac{RF + rd\|RD\ 1+gm}{(rd\|RD)} = \frac{10\ M\Omega+215\ k\Omega}{1+222\ mS\ (215\ k\Omega)}$ = **1.73 MΩ**

Zo = RF || rd || RD = 10 MΩ || 100 kΩ || 2.2 kΩ = **2.15 kΩ**

Av = -gm(RF || rd || RD) = -2.22 mS(2.15 kΩ) = **-4.77**

41. Determine Vo for the network of Fig. 9.67 if Vi = 20 mV.

Jawab :

$Vgs_q = 5.7\ V, g_m = 2k\ (Vgs_q - V_T) = 2(0.3 \times 10^{-3})(5.7\ V - 3.5\ V)$

$= 1.32\ mS$

$Rd = \frac{1}{30\mu S} = 33.33\ k\Omega$

$A_v = -\ g_m\ (R_F \| rd \| R_D) = -\ 1.32\ mS(22\ M\Omega \| 33.33\ K\Omega \| 10\ K\Omega)$

$= -\ 10.15$

$V_o = A_v\ V_i = (-10.15)(20\ mV) = -203\ mV$

42. Determine Vo for the network of Fig. 9.67 if Vi = 4 mV, $V_{GS(Th)} = 4$ V, and $ID_{(on)}$ = 4 mA, with

$V_{GS(on)} = 7$ V and $g_{or} = 20\ \mu S$.

Jawab =

$I_D = k\ (V_{GS} - V_T)^2$

$\therefore k = \frac{ID\ (on)}{(Vgs(on)-Vt)2} = \frac{4\ mA}{(7V-4V)2} = 0.444 \times 10^{-3}$

$g_m = 2k\ (Vgs_q - V_{GS(Th)}) = 2(0.444 \times 10^{-3})\ (7V - 4V)$

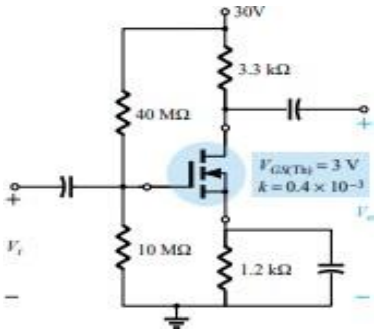
$= 2.66\ mS$

$A_v = -g_m(R_F \| r_d \| R_D) = -(2.66\ mS)(22\ M\Omega \| \underbrace{50\ k\Omega \| 10\ k\Omega}_{8.33\ k\Omega}) = -22.16$

$\cong 8.33\ k\Omega$

$V_o = A_v V_i = (-22.16)(4\ mV) = -88.64\ mV$

43. Determine the output voltage for the network of Fig. 9.68 if VI = 0.8 mV and rd = 40 kΩ.



Jawab :

$V_{GSQ} = 4.8$ V, $g_m = 2k\ (V_{GSQ} - V_{GS(Th)} = 2(0.4 \times 10^{-3})(4.8\ V - 3V) = 1.44\ mS$

$A_V = -g_m (rd \| R_D) = -(1.44\ mS)(40\ k\Omega) \| 3.3\ k\Omega) = -4.39$

$V_o = A_V A_i = (-4.39)(0.8\ mV) = -3.51\ mV$

44. Design the fixed-bias network of Fig. 9.69 to have a gain of 8

Jawab :

$rd = \frac{1}{Yon} = \frac{1}{20\ \mu S} = 50\ k\Omega$

$V_{GSq} = 0\ V, \therefore g_m = g_{m0} = \frac{21\ dss}{|Vp|} = \frac{2(8\ mA)}{2.5\ V} = 6.4\ mS$

$|A_V| = g_m (r_d \| R_D)$

$8 = (6.4\ mS)(50\ K\Omega \| R_D)$

$\frac{8}{6.4\ mS} = 1.25\ k\Omega = \frac{50\ K\Omega RD}{50\ K\Omega + RD}$

And $R_D = 1.28\ K\Omega$

Use $R_D = 1.3\ K\Omega$

45. Design the self-bias network of Fig. 9.70 to have a gain of 10. The device should be biased at

$V_{GSQ} = \frac{1}{3} V_P$.

Jawab :

$V_{GSq} = \frac{1}{3} V_P = \frac{1}{3}(-3\ V) = -1\ V$

$I_{D_Q} = I_{DSS}\left(1 - \frac{V_{GS_Q}}{V_P}\right)^2 = 12\ mA\left(1 - \frac{-1\ V}{-3\ V}\right)^2 = 5.33\ mA$

$R_S = \frac{Vs}{I\,Dq} = \frac{1V\,533}{mA} = 187.62\ \Omega \therefore$ Use $R_S = 180\ \Omega$

$g_m = \frac{21\ dss}{Vp}\left(1 - \frac{Vgsq}{Vp}\right) = \frac{2(12\ mA)}{3V}\left(1 - \frac{-1\ V}{-3\ V}\right) = 5.33\ mS$

$A_V = -g_m (R_D \| r_d) = -10$

Or $R_D \| 40\ K\Omega = \frac{-10}{533\ mS} = 1.876\ k\Omega$

$= 1.876\ k\ \Omega$

$40\ k\Omega R_D = 1.876\ k\Omega + 75.04\ k\Omega^2$

$38.124\ R_D = 75.04\ k\Omega$

$R_D = 1.97\ k\Omega \approx R_D = 2k\Omega$